

Forensic Investigation Report

Case ID	CASE_2CF24DBA5F	Report ID	RPT-2CF24DBA5F-C2E2DACB
Generated (UTC)	2026-07-25T14:45:18.979Z	Report version	v4

Every statement in this report references stored forensic findings; no content is generated outside computed results. Section "Report Provenance & Integrity" lists the SHA-256 digests of the source artifacts.

Executive Summary

- Case CASE_2CF24DBA5F contains 5 evidence item(s), each acquired under SHA-256 chain-of-custody verification.
- The weighted correlation engine found 10 related evidence pair(s) out of 10 analysed [correlation_analysis.json].
- The strongest suspect anchor is 'security@nabil-bank-alerts.com' (emails) with confidence 54/100 [suspect_assessment.json].
- Observed attack progression: initial_contact -> financial_transaction -> post_attack -> social_engineering -> credential_theft [timeline_analysis.json].

Scope & Methodology

- **objective:** Acquire, verify, correlate and reconstruct the digital evidence for this case, and derive investigative leads (suspect anchors, campaigns, cross-case links) strictly from stored, hash-verified artifacts.
- **evidence scope:** 5 evidence item(s) acquired through the CIIS intake pipeline under SHA-256 chain-of-custody control.
- **methodology:**
 - Phase 1 - Acquisition & OCR: PaddleOCR PP-OCRv5 text extraction with per-item confidence scoring; SHA-256 fingerprint recorded at intake and re-verified at read.
 - Phase 1 - Forensics: metadata/EXIF consistency, forgery signals, logo detection and evidence-confidence scoring stored per item under storage/forensics/.
 - Phase 2 - Correlation: weighted entity-overlap engine scoring every evidence pair; results in correlation_analysis.json.
 - Phase 2 - Cross-case: shared-entity matching against every other analysed case (cross_case_correlation.json).
 - Phase 2 - Campaigns / Suspects / Timeline: clustering, anchor derivation and event reconstruction over the correlated evidence set.
 - Threat intelligence: URL/domain indicators scored by the configured provider (static indicator file, or the trained phishing classifier when CIIS_ML_THREAT_INTEL=1); per-indicator results in 'Model Prediction Results'.
 - Reporting: this document is assembled exclusively from the stored outputs above; it contains no free-text generation.
- **reproducibility:** Re-running the analysis against the same stored evidence reproduces every figure herein; artifacts are versioned and never overwritten.

Case Overview

- **case id:** CASE_2CF24DBA5F
- **evidence count:** 5
- **first evidence:** 2026-07-25T14:22:38.355Z
- **last evidence:** 2026-07-25T14:24:07.592Z
- **file types:**
 - jpg
 - pdf
 - png
 - txt
 - url

Evidence Summary

Evidence	File	Acquired	OCR conf.	Integrity
EVID_00001	https_plasticostermoencogibles.pe_scss_css_scss_.url	2026-07-25T14:22:38	100%	VERIFIED
EVID_00002	police_scan_report.pdf	2026-07-25T14:23:23	100%	VERIFIED
EVID_00003	phishing_email_screenshot.png	2026-07-25T14:23:26	99%	VERIFIED
EVID_00004	const.jpg	2026-07-25T14:23:41	90%	VERIFIED
EVID_00005	whatsapp_chat_export.txt	2026-07-25T14:24:07	100%	VERIFIED

SHA-256 digests for each item are listed in the Appendix (chain of custody).

Correlation Analysis

- **pair count:** 10
- **related pair count:** 10
- **strength distribution:**
 - **WEAK:** 10
- **top relationships:**
 - **pair:** EVID_00001 <-> EVID_00002
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (https_plasticostermoencogibles.pe_scss_css_scss_.url) and EVID_00002 (police_scan_report.pdf) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (https_plasticostermoencogibles.pe_scss_css_scss_.url) and EVID_00003 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00004
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (https_plasticostermoencogibles.pe_scss_css_scss_.url) and EVID_00004 (const.jpg) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00005
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (https_plasticostermoencogibles.pe_scss_css_scss_.url) and EVID_00005 (whatsapp_chat_export.txt) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00002 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00002 (police_scan_report.pdf) and EVID_00003 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].

Cross-Case Correlation

No cross-case correlations were found for this case.

Campaign Analysis

- **campaign count:** 0
- **unclustered evidence:**
 - EVID_00001
 - EVID_00002
 - EVID_00003
 - EVID_00004
 - EVID_00005
- **campaigns:**
 - none

Timeline Analysis

- **summary:** 5 event(s) spanning 4853.2 hour(s); 0 timestamp(s) unresolved. Observed scam progression: initial_contact -> financial_transaction -> post_attack -> social_engineering -> credential_theft.
- **stage progression:**
 - initial_contact
 - financial_transaction
 - post_attack
 - social_engineering
 - credential_theft
- **progression consistent:** False
- **milestones:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** Investigation start - first reconstructed evidence event
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** First observation of stage 'initial_contact' (EVID_00005)
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** First observation of stage 'financial_transaction' (EVID_00005)
 - **timestamp:** 2026-07-25T14:23:23.722000+00:00
 - **description:** First observation of stage 'post_attack' (EVID_00002)
 - **timestamp:** 2026-07-25T14:23:26.295000+00:00
 - **description:** First observation of stage 'social_engineering' (EVID_00003)
 - **timestamp:** 2026-07-25T14:23:26.295000+00:00
 - **description:** First observation of stage 'credential_theft' (EVID_00003)
- **critical events:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **evidence id:** EVID_00005
 - **reasons:**
 - contains money entity/entities: NPR 2000
 - **timestamp:** 2026-07-25T14:23:23.722000+00:00
 - **evidence id:** EVID_00002
 - **reasons:**
 - contains money entity/entities: NPR 250000

Suspect Assessment

- **suspect id:** SUSPECT_CASE_2CF24DBA5F_01
- **identity:** emails:security@nabil-bank-alerts.com
- **confidence score:** 53.8
- **confidence level:** MODERATE
- **risk level:** MEDIUM
- **evidence ids:**
 - EVID_00003
- **explanation:** Suspect anchor 'security@nabil-bank-alerts.com' (emails) scores 53.8/100 (MODERATE, risk MEDIUM) across 1 evidence item(s): EVID_00003. Identity strength: 75/100 (weight 0.25) -

'security@nabil-bank-alerts.com' is a email - identity weight 75/100 for this anchor type. Evidence count: 25/100 (weight 0.20) - appears in 1 of 5 evidence item(s). Evidence confidence: 50/100 (weight 0.15) - mean Phase-1 evidence confidence of its evidence set is 50/100 (no Phase-1 scores stored; neutral 50 assumed). Threat intelligence: 100/100 (weight 0.15) - co-occurs with threat-flagged indicator 'http://nabil-verify.example-scam.top/login' in EVID_00003. Correlation strength: 50/100 (weight 0.15) - fewer than two evidence items (or no correlation input); neutral 50. Timeline span: 0/100 (weight 0.10) - activity spans 0.0 day(s) across its evidence set.

Threat Intelligence Summary

- intel available: 1.0
- indicators checked: 5.0
- malicious indicators: 3.0
- evidence with threats: 1.0
- threat evidence ratio: 0.2

Model Prediction Results

Indicator	Verdict	Risk	Confidence	Source model
http://nabil-verify.example-scam.top/login	MALICIOUS	81	81%	ml:xgboost
nabil-bank-alerts.com	MALICIOUS	79	79%	ml:xgboost
https://plasticostermoencogibles.pe/scss/css/scss/	SUSPICIOUS	42	94%	ml:xgboost

http://nabil-verify.example-scam.top/login

- domain: nabil-verify.example-scam.top · SSL: UNKNOWN · SPF: missing · DMARC: missing · trust: 35/100
- The hybrid decision engine confirmed this URL is phishing based on agreement between the ML model and threat indicators.
 - AI model classified this URL as phishing with 100% confidence.
 - The top-level domain is frequently associated with phishing and spam campaigns (risk score: 0.7).
 - SSL status could not be verified.
 - The URL does not use HTTPS, meaning data is transmitted without encryption.
 - The URL contains suspicious keyword(s) associated with phishing.

nabil-bank-alerts.com

- domain: nabil-bank-alerts.com · SSL: UNKNOWN · SPF: missing · DMARC: missing · trust: 50/100
- The hybrid decision engine confirmed this URL is phishing based on agreement between the ML model and threat indicators.
 - AI model classified this URL as phishing with 100% confidence.
 - SSL status could not be verified.
 - The URL does not use HTTPS, meaning data is transmitted without encryption.
 - The URL contains suspicious keyword(s) associated with phishing.
 - The domain is missing SPF and DMARC email-authentication record(s), which legitimate organisations typically configure.

https://plasticostermoencogibles.pe/scss/css/scss/

- domain: plasticostermoencogibles.pe · resolves to: 75.102.22.119 · registrar: Yachay Telecomunicaciones · hosted by: DEFT.COM, United States · SSL: VALID (35 days left) · SPF: present · DMARC: missing · trust: 75/100
- The hybrid decision engine flagged this URL as suspicious due to mixed signals between the ML model, active rules, and threat intelligence.
 - AI model classified this URL as phishing with 100% confidence.
 - The domain is missing DMARC email-authentication record(s), which legitimate organisations typically configure.
 - indicators classified: 3
 - flagged malicious: 2

Evidence Quality Summary

- mean image quality: 0.0
- mean evidence confidence: 0.0
- mean forgery score: 0.0
- max forgery score: 0.0
- mean ocr confidence: 0.98
- hash verified count: 5.0

Metadata Summary

- **note:** No Phase-1 metadata reports stored for this case.

Investigation Statistics

- **entity statistics:**
 - dates: 1
 - domains: 3
 - emails: 1
 - money: 2
 - times: 3
 - urls: 2
- **campaign statistics:**
 - campaign count: 0.0
 - largest campaign size: 0.0
 - clustered evidence: 0.0
 - unclustered evidence: 5.0
 - mean campaign confidence: 0.0
- **timeline statistics:**
 - event count: 5.0
 - resolved event count: 5.0
 - unresolved event count: 0.0
 - inferred event count: 4.0
 - stage count: 5.0
 - critical event count: 2.0
 - timeline span hours: 4853.19
- **correlation statistics:**
 - pair count: 10.0
 - related pair count: 10.0
 - mean confidence: 0.2212
 - max confidence: 0.2212

Confidence Analysis

- **note:** No Phase-1 confidence scores stored for this case.

Investigation Conclusion

- 5/5 evidence item(s) passed SHA-256 chain-of-custody verification.
- The evidence set is internally connected (10 weighted relationship(s)), consistent with related activity rather than isolated incidents.
- Investigation should focus on anchor 'security@nabil-bank-alerts.com' (54/100 confidence).
- Observed stage order deviates from the canonical scam sequence; evidence acquisition order should be reviewed.

Recommendations

- Pursue subscriber/KYC records for email 'security@nabil-bank-alerts.com' (suspect confidence 54/100).
- Prioritise the 2 critical event(s) involving OTP/financial entities for victim-impact assessment.

- Case priority: LOW (21.6/100) - Low urgency: archive-ready unless new evidence raises the correlation or threat picture.

Report Provenance & Integrity

- **report id:** RPT-2CF24DBA5F-C2E2DACB
- **generated at:** 2026-07-25T14:45:18.979Z
- **generator:** CIIS Phase-2 reporting module (template-over-data; no free-text generation)
- **evidence set digest:** f8320f982f3c29e5b76842e5e4ffef335f82140f5388e98ff6a8c8f992f16825
- **evidence set digest note:** SHA-256 over the sorted SHA-256 digests of every evidence item; any change to the evidence set changes this value.

Source artifact digests (SHA-256):

correlation_analysis_v9.json: ca426761da8574f896a033abc3d929563c4a5c12da58e63302e865bbcb15b9a1
cross_case_correlation_v3.json: c62e8db800905c481bc8f84d740e477ef15811f557e91e791318f20f69514027
campaign_analysis_v4.json: 867165c44b0d22d55f575534b71edba21658de1fc48dfb6006ada9d078798f0b
suspect_assessment_v4.json: 6bd8090d54379234f54a025b6669a4471c5fb99ce1da55142ef5664dafd2b6a2
timeline_analysis_v9.json: 4d97b7f30799f982f26c58a57ebd602e39b593b7bd1c23fc5090ef2c510a28fd
analytics_v4.json: ee5dfd0f7d535152fc230ace2ecd580341159d5e88371b5eb26f5a2f21f8195d
case_priority_v4.json: 33a2d5f473a69565f32cb5e83f86df9c708e1ef79c607e8f806296be6fa39c22
graph_v9.json: 5c8e861eb531efca38b68454631b286867c1796a137f2572de8cd83f0b5265cd

Appendix

- **chain of custody:**
 - **evidence id:** EVID_00001
 - **sha256:** fab44e99bd90f1ffe214d4d2bb6580e4c0d52f5fe930e773d96cf92ee820a2f5
 - **upload time:** 2026-07-25T14:22:38.355Z
 - **status:** processed
 - **evidence id:** EVID_00002
 - **sha256:** 63b677916b711ece64fc71292577dfb70a2385c2de817cd0200f86d777b0b350
 - **upload time:** 2026-07-25T14:23:23.722Z
 - **status:** processed
 - **evidence id:** EVID_00003
 - **sha256:** 8ef0f27a84960b656d9051892e36096ce56fe9f30eb9a935d4d4330a002b9b7f
 - **upload time:** 2026-07-25T14:23:26.295Z
 - **status:** processed
 - **evidence id:** EVID_00004
 - **sha256:** d07bcc76b5d280385a06f106c083edc33bff69e82ac63fddf8c433f6b4a5fc02
 - **upload time:** 2026-07-25T14:23:41.798Z
 - **status:** processed
 - **evidence id:** EVID_00005
 - **sha256:** c7ab347cfbd2b80c944c85b3442f3b2c400da1d3b92962f9ded8598e1e06cfb2
 - **upload time:** 2026-07-25T14:24:07.592Z
 - **status:** processed
- **stored artifacts root:** storage/investigation/<CASE_ID>/
- **phase1 artifacts root:** storage/forensics/<EVIDENCE_ID>/

Prepared by (system)

Cybercrime Investigation Intelligence System
Automated Phase-2 reporting module

Reviewed by (investigator)

Date:

Date:

Signature: